

Theoretical, Permutation, and Empirical Null Distributions

In classical significance testing, the null distribution plays the role of devil's advocate: a standard that the observed data must exceed in order to convince the scientific world that something interesting has occurred. We observe, say, $z = 2$, and note that in a hypothetical "long run" of observations from a $\mathcal{N}(0, 1)$ distribution less than 2.5% of the draws would exceed 2, thereby discrediting the uninteresting null distribution as an explanation.

Considerable effort has been expended trying to maintain the classical model in large-scale testing situations, as seen in Chapter 3, but there are important differences that affect the role of the null distribution when the number of cases N is large:

- With $N = 10\,000$ for example, the statistician has his or her own "long run" in hand. This diminishes the importance of theoretical null calculations based on mathematical models. In particular, it may become clear that the classical null distribution appropriate for a single-test application is in fact wrong for the current situation.
- Scientific applications of single-test theory most often suppose, or hope for, *rejection* of the null hypothesis, perhaps with power = 0.80. Large-scale studies are usually carried out with the expectation that most of the N cases will *accept* the null hypothesis, leaving only a small number of interesting prospects for more intensive investigation.
- Sharp null hypotheses, such as $H_0 : \mu = 0$ for $z \sim \mathcal{N}(\mu, 1)$, are less important in large-scale studies. It may become clear that most of the N cases have small, uninteresting but non-zero values of μ , leaving just a few genuinely interesting cases to identify. As we will discuss, this results in a broadening of classical null hypotheses.
- Large-scale studies allow empirical Bayes analyses, where the null distribution is put into a probabilistic context with its non-null competitors (as seen in Chapters 4 and 5).
- The line between estimation and testing is blurred in large-scale studies.

Large N isn't infinity and empirical Bayes isn't Bayes, so estimation efficiency of the sort illustrated in Figure 5.3 plays a major role in large-scale testing.

The theoretical null distribution provides a reasonably good fit for the prostate and DTI examples of Figure 5.1a and Figure 5.1b. This is a less-than-usual occurrence in my experience. A set of four large-scale studies are presented next in which the theoretical null has obvious troubles.¹ We will use them as trial cases for the more flexible methodology discussed in this chapter.

6.1 Four Examples

Figure 6.1 displays z -values for four large-scale testing studies, in each of which the theoretical null distribution is incompatible with the observed data. (A fifth, artificial, example ends the section.) Each panel displays the following information:

- the number of cases N and the histogram of the N z -values;
- the estimate $\hat{\pi}_{00}$ of the null proportion π_0 (2.7) obtained as in (4.46), (4.48) with $\alpha_0 = 0.5$, using the theoretical null density $f_0(z) \sim \mathcal{N}(0, 1)$;
- estimates $(\hat{\delta}_0, \hat{\sigma}_0, \hat{\pi}_0)$ for π_0 and the empirical null density $\hat{f}_0(z) \sim \mathcal{N}(\hat{\delta}_0, \hat{\pi}_0)$ obtained by the MLE method of Section 6.3, providing a normal curve fit to the central histogram;
- a heavy solid curve showing the empirical null density, scaled to have area $\hat{\pi}_0$ times that of the histogram (i.e., $Nd\hat{\pi}_0 \cdot \hat{f}_0(z)$, where d is the bin width);
- a light dotted curve proportional to $\hat{\pi}_{00} \cdot \varphi(z)$ ($\varphi(z) = \exp\{-\frac{1}{2}z^2\} / \sqrt{2\pi}$);
- small triangles on the x -axis indicating values at which the local false discovery rate $\widehat{\text{fdr}}(z) = \hat{\pi}_0 \hat{f}_0(z) / \hat{f}(z)$ based on the empirical null equals 0.2 (with $\hat{f}(z)$ as described in Section 5.2 using a natural spline basis with $J = 7$ degrees of freedom);
- small hash marks below the x -axis indicating z -values with $\widehat{\text{fdr}}(z_i) \leq 0.2$;
- the number of cases for which $\widehat{\text{fdr}}(z_i) \leq 0.2$.

What follows are descriptions of the four studies.

¹ Questions about the proper choice of a null distribution are not restricted to large-scale studies. They arise prominently in analysis of variance applications, for instance in whether to use the interaction or residual sums of squares for testing main effects in a two-way replicated layout.

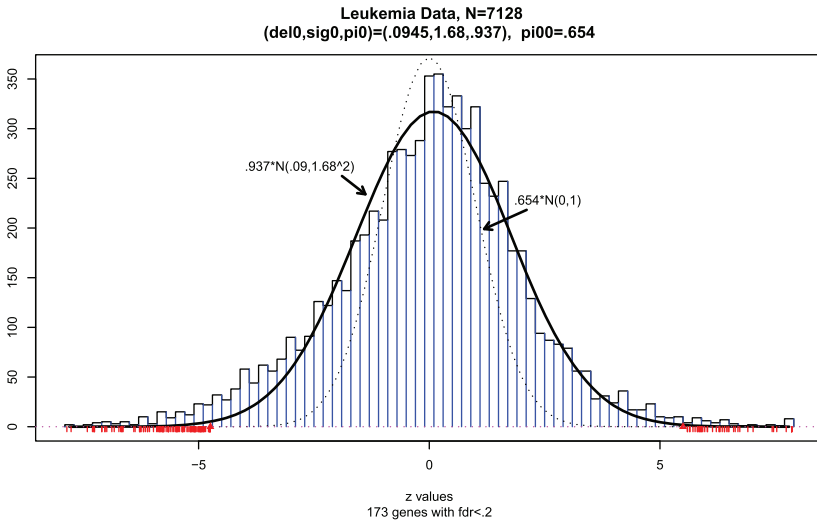


Figure 6.1a z -value histogram for leukemia study. Solid curve is empirical null, dotted curve theoretical $N(0, 1)$ null. Hash marks indicate z -values having $\widehat{\text{fdr}} \leq 0.2$. The theoretical null greatly underestimates the width of the central histogram.

A. Leukemia study

High-density oligonucleotide microarrays provided expression levels on $N = 7128$ genes for $n = 72$ patients, $n_1 = 45$ with ALL (acute lymphoblastic leukemia) and $n_2 = 27$ with AML (acute myeloid leukemia); the latter has the worse prognosis. The raw expression levels on each microarray, X_{ij} for gene i on array j , were transformed to a *normal scores* value

$$x_{ij} = \Phi^{-1} \left(\frac{\text{rank}(X_{ij}) - 0.5}{N} \right), \quad (6.1)$$

$\text{rank}(X_{ij})$ being the rank of X_{ij} among the N raw scores on array j , and Φ the standard normal cdf. This was necessary to eliminate response disparities among the n microarrays as well as some wild outlying values.² Z -values z_i were then obtained from two-sample t -tests comparing AML with ALL patients as in (2.1)–(2.5), now with 70 degrees of freedom.

We see that the z -value histogram is highly overdispersed compared to a $N(0, 1)$ theoretical null. The empirical null is $N(0.09, 1.68^2)$ with $\hat{\pi}_0 = 0.937$; 173 of the 7128 genes had $\widehat{\text{fdr}}(z) \leq 0.20$. If we insist on using the

² Some form of standardization is almost always necessary in microarray studies.

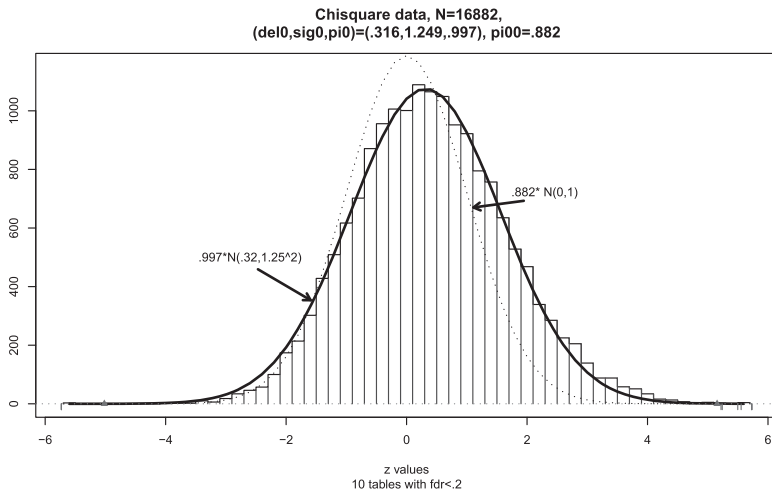


Figure 6.1b z -value histogram for chi-square study as in Figure 6.1a. Here the theoretical null is mis-centered as well as too narrow.

theoretical null, $\hat{\pi}_{00}$ is estimated to be only 0.654, while 1548 of the genes now have $\text{fdr}(z_i) \leq 0.20$. Perhaps it is possible that 2464 ($= (1 - \hat{\pi}_{00}) \cdot N$) of the genes display AML/ALL genetic differences, but it seems more likely that there is something inappropriate about the theoretical null. Just what might go wrong is the subject of Section 6.4.

B. Chi-square data

This experiment studied the binding of certain chemical tags at sites within $N = 16\,882$ genes. The number K of sites per gene ranged from three up to several hundred, median $K = 12$. At each site within each gene the number of bound tags was counted. The count was performed under two different experimental conditions, with the goal of the study being to identify genes where the proportion of tags differed between the two conditions. Table 6.1 shows the $K \times 2$ table of counts for the first of the genes, in which $K = 8$.

A z -value z_i was calculated for table _{i} as follows:

- (i) One count was added to each entry of table _{i} .
- (ii) S_i , the usual chi-square test statistic for independence, was computed for the augmented table.

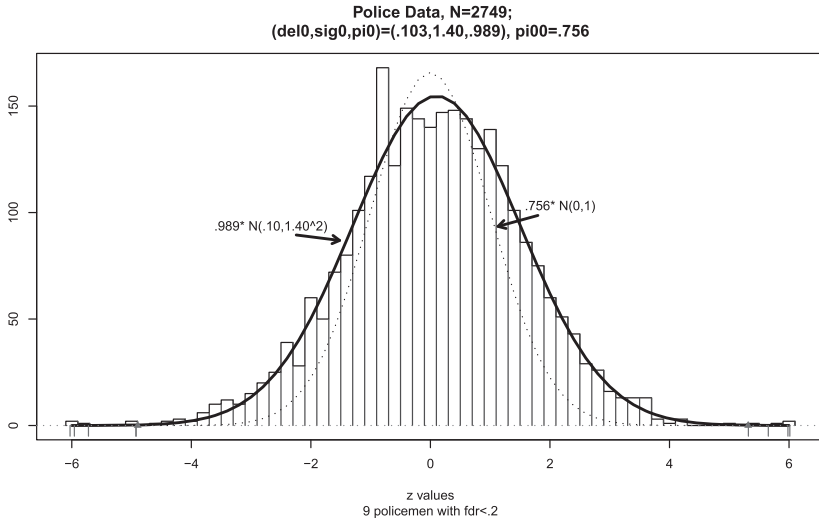


Figure 6.1c z-value histogram for police data as in Figure 6.1a. The theoretical null is about 40% too narrow.

Table 6.1 First of $N = 16\,882$ $K \times 2$ tables for the chi-square data; shown are the number of tags counted at each site under two different experimental conditions.

Site	1	2	3	4	5	6	7	8
# condition 1	8	8	4	2	1	5	27	9
# condition 2	5	7	1	0	11	4	4	10

(iii) An approximate p -value was calculated,

$$p_i = 1 - F_{K-1}(S_i) \quad (6.2)$$

where F_{K-1} was the cdf of a standard chi-square distribution having $K-1$ degrees of freedom.

(iv) The assigned z -value for table _{i} was

$$z_i = \Phi^{-1}(1 - p_i) \quad (6.3)$$

with small values of p_i corresponding to large z_i . For table₁ in Table 6.1, $p_1 = 0.00132$ and $z_1 = 3.01$.

Exercise 6.1 Verify this calculation.

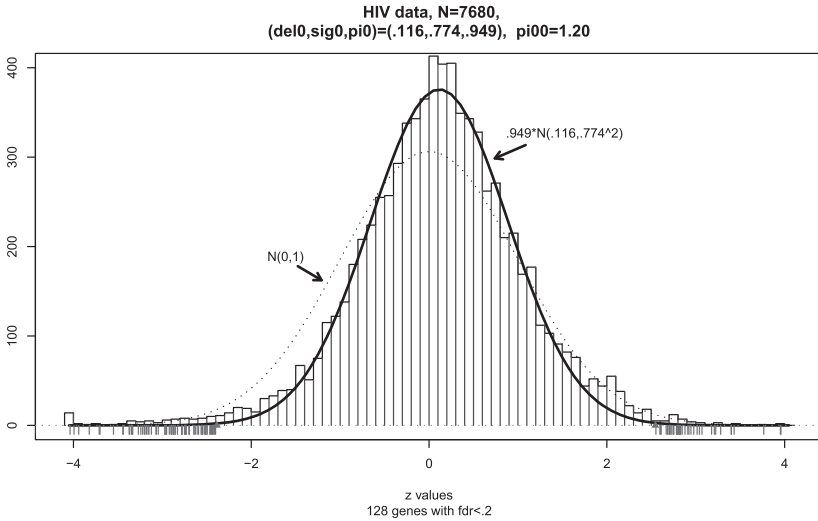


Figure 6.1d z -value histogram for HIV data as in Figure 6.1a. Note reduced scale on x -axis. In this case the theoretical null is too wide.

The ad hoc addition of one count in step (i) greatly stabilized the histogram of z -values.³ Our methods do not require classical forms for the test statistics such as the standard chi-square definition, but they do depend on being able to approximate the center of the z -value histogram with a normal curve. This leads to questions of comparability and relevance in simultaneous inference, discussed further in Chapter 10.

The empirical null is $\mathcal{N}(0.32, 1.25^2)$ so that, besides being underdispersed, the theoretical null is mis-centered, as apparent in Figure 6.1b. Only ten of the tables had $\widehat{\text{fdr}}(z_i) \leq 0.2$, seven on the right and three on the left. If anything, we would expect the chi-square statistic S_i to be too big instead of too small, but that is not the case here. The three tables on the left with $\widehat{\text{fdr}} \leq 0.2$ all had $K = 3$, the smallest value, raising further doubts about their selection. Chapter 10 returns to this example in its discussion of comparability.

³ The SAM program of Section 4.5 employs a similar tactic in two-sample comparisons: a small constant is added to the denominator of the usual two-sample t -statistic in order to prevent low-variance cases from dominating the tails of the z -value distribution.

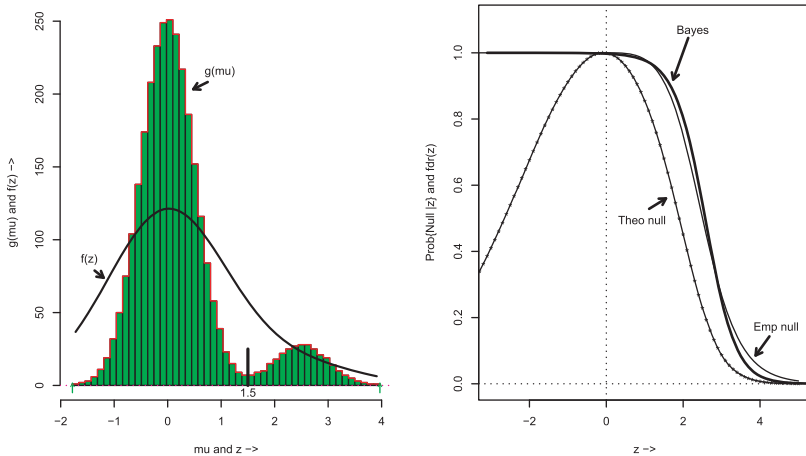


Figure 6.1e *Left panel:* Histogram shows $N = 3000$ draws from prior density $g(\mu)$ (6.6); curve $f(z)$ is corresponding density of observations $z \sim N(\mu, 1)$. *Right panel:* Heavy curve is Bayes posterior probability $\Pr\{\mu \leq 1.5|z\}$; it is well approximated by the empirical null estimate $\widehat{\text{fdr}}(z)$ (6.9), light curve; $\text{fdr}(z)$ based on the theoretical null $f_0 \sim N(0, 1)$ is a poor match to the Bayes curve.

C. Police data

A study of possible racial bias in police pedestrian stops was conducted in New York City in 2006. Each of $N = 2749$ officers was assigned a score z_i on the basis of their stop data, with large positive values of z_i being possible evidence of bias. In computing z_i , an ingenious two-stage logistic regression analysis was used to compensate for differences in the time, place, and context of the individual stops.

Let $\mathbf{x}_{ij}$ represent the vector of covariates for officer i , stop j . A greatly oversimplified version of the logistic regression model actually used is

$$\text{logit}\{\Pr(y_{ij} = 1)\} = \beta_i + \gamma' \mathbf{x}_{ij} \quad (6.4)$$

where y_{ij} indicates whether or not the stopped person was a member of a defined minority group, β_i is the “officer effect,” and γ is the vector of logistic regression coefficient for the covariates. The z -score for officer i was

$$z_i = \hat{\beta}_i / \text{se}(\hat{\beta}_i) \quad (6.5)$$

where $\hat{\beta}_i$ and $\text{se}(\hat{\beta}_i)$ are the usual estimate and approximate standard error for β_i .

A standard analysis would rely on the theoretical null hypothesis $z_i \sim \mathcal{N}(0, 1)$. However, the histogram of all $N = 2749$ z_i values is much wider, the empirical null being $\hat{f}_0 \sim \mathcal{N}(0.10, 1.40^2)$. This resulted in nine officers having $\text{fdr}(z_i) \leq 0.20$, only four of whom were on the right (i.e., “racially biased”) side. The estimated non-null proportion was $\hat{\pi}_1 = 0.011$, only about 1/5 of which applied to the right side according to the smoothed non-null counts (5.41).

There is a lot at stake here. Relying on the theoretical $\mathcal{N}(0, 1)$ null gives $\hat{\pi}_1 = 0.24$, more than 20 times greater and yielding 122 officers having positive z -scores with $\text{fdr}(z_i) \leq 0.20$. The argument for empirical null analysis says we should judge the extreme z -scores by comparison with central variability of the histogram and not according to a theoretical standard. Section 6.4 provides practical arguments for doubting the theoretical standard.

D. HIV data

This was a small study in which $n_2 = 4$ HIV-positive subjects were compared with $n_1 = 4$ healthy controls using cDNA microarrays that measured expression levels for $N = 7680$ genes. Two-sample t -tests (on the logged expressions) yielded z -values z_i as in (2.2)–(2.5) except now with 6 rather than 100 degrees of freedom.

Unlike all of our previous examples (including the prostate and DTI studies), here the central histogram is *less* dispersed than a theoretical $\mathcal{N}(0, 1)$ null, with $\hat{f}_0(z) \sim \mathcal{N}(0.12, 0.77^2)$ and $\hat{\pi}_0 = 0.949$. (Underdispersion makes the theoretical null estimate equal the impossible value $\hat{\pi}_{00} = 1.20$.) Using the theoretical null rather than the empirical null reduces the number of genes having $\text{fdr} \leq 0.2$ from 128 to 20.

Figure 6.1e concerns an artificial example involving an overdispersed null distribution, similar to the leukemia, chi-square, and police situations. Pairs (μ_i, z_i) have been independently generated according to the Bayesian hierarchical model (2.47), $\mu \sim g(\cdot)$ and $z|\mu \sim \mathcal{N}(\mu, 1)$; the prior density $g(\mu)$, represented as a histogram in the left panel, is bimodal,

$$g(\mu) = 0.9 \cdot \varphi_{0,0.5}(\mu) + 0.1 \cdot \varphi_{2.5,0.5}(\mu) \quad (6.6)$$

where $\varphi_{a,b}$ represents the density of a $\mathcal{N}(a, b^2)$ distribution. However, the mixture density $f(z) = \int g(\mu)\varphi(z - \mu)d\mu$ is unimodal, reflecting the secondary mode of $g(\mu)$ only by a heavy right tail.

A large majority of the true effects μ generated from (6.6) will have

small uninteresting values centered at, but not exactly equaling, zero. The interesting cases, those with large μ , will be centered around 2.5. Having observed z , we would like to predict whether the unobserved μ is interesting or uninteresting. There is no sharp null hypothesis here, but the shape of $g(\mu)$ suggests defining

$$\text{Uninteresting: } \mu \leq 1.5 \quad \text{and} \quad \text{Interesting: } \mu > 1.5. \quad (6.7)$$

The heavy curve in the right panel is the Bayes prediction rule for “uninteresting,”

$$\Pr\{\mu \leq 1.5|z\}. \quad (6.8)$$

This assumes that prior density $g(\mu)$ is known to the statistician. But what if it isn't? The curve marked “Emp null” is the estimated local false discovery rate

$$\widehat{\text{fdr}}(z) = \hat{\pi}_0 \hat{f}_0(z) / \hat{f}(z) \quad (6.9)$$

obtained using the central matching method of Section 6.2 which gave empirical null estimates

$$\hat{\pi}_0 = 0.93 \quad \text{and} \quad \hat{f}_0(z) \sim \mathcal{N}(0.02, 1.14^2) \quad (6.10)$$

based on $N = 3000$ observations z_i . We see that it nicely matches the Bayes prediction rule, even though it did not require knowledge of $g(\mu)$ or of the cutoff point 1.5 in (6.7).

The point here is that empirical null false discovery rate methods can deal with “blurry” null hypotheses, in which the uninteresting cases are allowed to deviate somewhat from a sharp theoretical null formulation. (Section 6.4 lists several reasons this can happen.) The empirical null calculation absorbs the blur into the definition of “null”. Theoretical or permutation nulls fail in such situations, as shown by the beaded curve in the right panel.

6.2 Empirical Null Estimation

A null distribution is not something one estimates in classic hypothesis testing theory:⁴ theoretical calculations provide the null, which the statistician must use for better or worse. Large-scale studies such as the four examples in Section 6.1 can cast severe doubt on the adequacy of the theoretical null. Empirical nulls, illustrated in the figures, use each study's own data to estimate an appropriate null distribution. This sounds circular, but isn't. We

⁴ An exception arising in permutation test calculations is discussed in Section 6.5.

will see that a key assumption for empirically estimating the null is that π_0 , the proportion of null cases in (2.7), is large, say

$$\pi_0 \geq 0.90 \quad (6.11)$$

as in (2.8), allowing the null distribution opportunity to show itself.

The appropriate choice of null distribution is not a matter of local fdr versus tail area Fdr: both are equally affected by an incorrect choice. Nor is it a matter of parametric versus non-parametric procedures. Replacing t -statistics with Wilcoxon test statistics (each scaled to have mean 0 and variance 1 under the usual null assumptions) gives empirical null $\hat{f}_0(z) \sim \mathcal{N}(0.12, 1.72^2)$ for the leukemia data, almost the same as in Figure 6.1a.

The two-groups model (2.7) is unidentifiable: a portion of $f_1(z)$ can be redefined as belonging to $f_0(z)$ with a corresponding increase in π_0 . The *zero assumption* (4.44), that the non-null density $f_1(z)$ is zero near $z = 0$, restores identifiability and allows estimation of $f_0(z)$ and π_0 from the central histogram counts.

Exercise 6.2 Suppose $\pi_0 = 0.95$, $f_0 \sim \mathcal{N}(0, 1)$, and f_1 is an equal mixture of $\mathcal{N}(2.5, 1)$ and $\mathcal{N}(-2.5, 1)$. If we redefine the situation to make (4.44) true in $\mathcal{A}_0 = [-1, 1]$, what are the new values of π_0 and $f_0(z)$?

Define

$$f_{\pi_0}(z) = \pi_0 f_0(z) \quad (6.12)$$

so that

$$\text{fdr}(z) = f_{\pi_0}(z)/f(z), \quad (6.13)$$

(5.2). We assume that $f_0(z)$ is normal but not necessarily $\mathcal{N}(0, 1)$, say

$$f_0(z) \sim \mathcal{N}(\delta_0, \sigma_0^2). \quad (6.14)$$

This yields

$$\log(f_{\pi_0}(z)) = \left[\log(\pi_0) - \frac{1}{2} \left\{ \frac{\delta_0^2}{\sigma_0^2} + \log(2\pi\sigma_0^2) \right\} \right] + \frac{\delta_0}{\sigma_0^2} z - \frac{1}{2\sigma_0^2} z^2, \quad (6.15)$$

a quadratic function of z .

Central matching estimates $f_0(z)$ and π_0 by assuming that $\log(f(z))$ is quadratic near $z = 0$ (and equal to $f_{\pi_0}(z)$),

$$\log(f(z)) \doteq \beta_0 + \beta_1 z + \beta_2 z^2, \quad (6.16)$$

estimating $(\beta_0, \beta_1, \beta_2)$ from the histogram counts y_k (5.12) around $z = 0$ and

matching coefficients between (6.15) and (6.16): yielding $\sigma_0^2 = -1/(2\beta_2)$ for instance. Note that a different picture of matching appears in Chapter 11.

Exercise 6.3 What are the expressions for δ_0 and π_0 ?

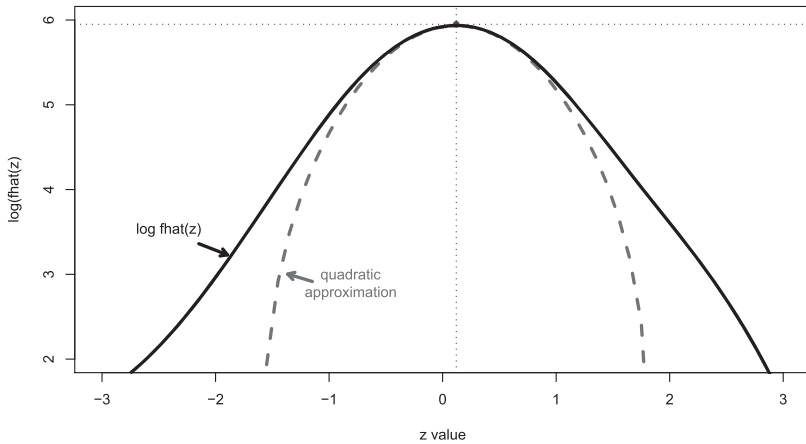


Figure 6.2 Central matching estimation of $f_0(z)$ and π_0 for the HIV data of Section 6.1, (6.15)–(6.16): $(\hat{\delta}_0, \hat{\sigma}_0, \hat{\pi}_0) = (0.12, 0.75, 0.93)$. Chapter 11 gives a different geometrical picture of the matching process.

Figure 6.2 shows central matching applied to the HIV data of Section 6.1. The quadratic approximation $\log \hat{f}_{\pi_0}(z) = \hat{\beta}_0 + \hat{\beta}_1 z + \hat{\beta}_2 z^2$ was computed as the least square fit to $\log \hat{f}(z)$ over the central 50% of the z -values (with the calculation discretized as in Section 5.2), yielding

$$(\hat{\delta}_0, \hat{\sigma}_0, \hat{\pi}_0) = (0.12, 0.75, 0.93). \quad (6.17)$$

These differ slightly from the values reported in Section 6.1 because those were calculated using the MLE method of Section 6.3.

The zero assumption (4.44) is unlikely to be literally true in actual applications. Nevertheless, central matching tends to produce nearly unbiased estimates of $f_0(z)$, at least under conditions (2.47), (2.49):

$$\begin{aligned} \mu &\sim g(\cdot) \quad \text{and} \quad z|\mu \sim \mathcal{N}(\mu, 1), \\ g(\mu) &= \pi_0 I_0(\mu) + \pi_1 g_1(\mu). \end{aligned} \quad (6.18)$$

Here $f_0(z) = \varphi(z)$, the $\mathcal{N}(0, 1)$ density, but we are not assuming that the

non-null density

$$f_1(z) = \int_{-\infty}^{\infty} g_1(\mu)\varphi(z - \mu) d\mu \quad (6.19)$$

satisfies the zero assumption. This introduces some bias into the central matching assessment of δ_0 and σ_0 but, as it turns out, not very much as long as $\pi_1 = 1 - \pi_0$ is small.

With π_0 fixed, let (δ_g, σ_g) indicate the value of (δ_0, σ_0) obtained under model (6.18) by an idealized version of central matching based on $f(z) = \int_{-\infty}^{\infty} g(\mu)\varphi(z - \mu)d\mu$,

$$\delta_g = \arg \max \{f(z)\} \quad \text{and} \quad \sigma_g = \left[-\frac{d^2}{dz^2} \log f(z) \right]_{\delta_g}^{-\frac{1}{2}}. \quad (6.20)$$

Exercise 6.4 What values of $\beta_0, \beta_1, \beta_2$ are implied by (6.20)? In what sense is (6.20) an idealized version of central matching?

We can ask how far (δ_g, σ_g) deviates from the actual parameters $(\delta_0, \sigma_0) = (0, 1)$ for $g(\mu)$ in (6.18). For a given choice of π_0 , let

$$\delta_{\max} = \max\{|\delta_g|\} \quad \text{and} \quad \sigma_{\max} = \max\{\sigma_g\}, \quad (6.21)$$

the maxima being over the choice of g_1 in (6.18). Table 6.2 shows the answers. In particular, for $\pi_0 \geq 0.90$ we always have

$$|\delta_g| \leq 0.07 \quad \text{and} \quad \sigma_g \leq 1.04 \quad (6.22)$$

so, no matter how the non-null cases are distributed, the idealized central matching estimates won't be badly biased.

Exercise 6.5 Show that $\sigma_g \geq 1$ under model (6.18).

Table 6.2 *Worst case values $\delta_{\max}$ and $\sigma_{\max}$ (6.21) as a function of $\pi_1 = 1 - \pi_0$.*

$\pi_1 = 1 - \pi_0$	.05	.10	.20	.30
$\sigma_{\max}$	1.02	1.04	1.11	1.22
$\delta_{\max}$	.03	.07	.15	.27

Figure 6.3 graphs $\delta_{\max}$ and $\sigma_{\max}$ as a function of $\pi_1 = 1 - \pi_0$. In addition to the general model (6.18), which provided the numbers in Table 6.2, $\sigma_{\max}$ is also graphed for the restricted version of (6.18) in which $g_1(\mu)$ is required

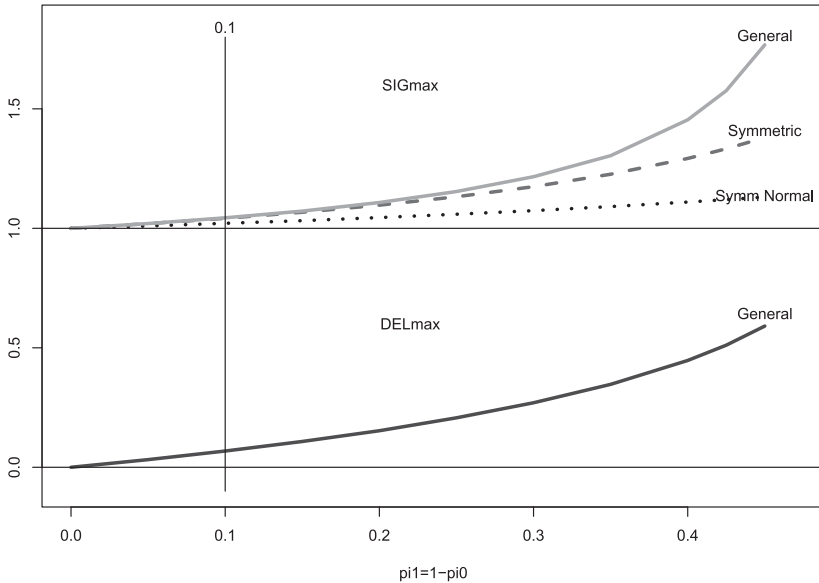


Figure 6.3 $\delta_{\max}$ and $\sigma_{\max}$ (6.21) as a function of $\pi_1 = 1 - \pi_0$; *Symmetric* restricts $g_1(\mu)$ in (6.18) to be symmetric around $\mu = 0$ and similarly *Symm Normal*.

to be symmetric about $\mu = 0$, and also the more restrictive version in which g_1 is both symmetric and normal. The worst case values in Table 6.2 have $g_1(\mu)$ supported on a single point. For example, $\sigma_{\max} = 1.04$ is achieved with $g_1(\mu)$ supported at $\mu_1 = 1.47$.

The default option in *locfdr* is the *MLE method*, discussed in the next section, not central matching. Slight irregularities in the central histogram, as seen in Figure 6.1a, can derail central matching. The MLE method is more stable, but pays the price of possibly increased bias.

Here is a derivation of $\sigma_{\max}$ for the symmetric case in Figure 6.3, the derivations for the other cases being similar. For convenience we consider discrete distributions $g(\mu)$ in (6.18), putting probability π_0 on $\mu = 0$ and π_j on pairs $(-\mu_j, \mu_j)$, $j = 1, 2, \dots, J$, so the mixture density equals

$$f(z) = \pi_0 \varphi(z) + \sum_{j=1}^J \pi_j [\varphi(z - \mu_j) + \varphi(z + \mu_j)] / 2. \quad (6.23)$$

Then we can take $\delta_j = 0$ in (6.20) by symmetry (0 being the actual maximum if π_0 exceeds 0.50). We consider π_0 fixed in what follows.

Defining $c_0 = \pi_0/(1 - \pi_0)$, $r_j = \pi_j/\pi_0$, and $r_+ = \sum_1^J r_j$, we can express σ_g in (6.20) as

$$\sigma_g = (1 - Q)^{-\frac{1}{2}} \quad \text{where } Q = \frac{\sum_1^J r_j \mu_j^2 e^{-\mu_j^2/2}}{c_0 r_+ + \sum_1^J r_j e^{-\mu_j^2/2}}. \quad (6.24)$$

Actually, $r_+ = 1/c_0$ and $c_0 r_+ = 1$, but the form in (6.24) allows unconstrained maximization of Q (and of σ_g) as a function of $\mathbf{r} = (r_1, r_2, \dots, r_J)$, subject only to $r_j \geq 0$ for $j = 1, 2, \dots, J$.

Exercise 6.6 Verify (6.24).

Differentiation gives

$$\frac{\partial Q}{\partial r_j} = \frac{1}{\text{den}} \left[\mu_j^2 e^{-\mu_j^2/2} - Q \cdot (c_0 + e^{-\mu_j^2/2}) \right] \quad (6.25)$$

with den the denominator of Q in (6.24). At a maximizing point $\mathbf{r}$ we must have

$$\frac{\partial Q(\mathbf{r})}{\partial r_j} \leq 0 \quad \text{with equality if } r_j > 0. \quad (6.26)$$

Defining $R_j = \mu_j^2/(1 + c_0 e^{\mu_j^2/2})$, (6.25)–(6.26) give

$$Q(\mathbf{r}) \geq R_j \quad \text{with equality if } r_j > 0. \quad (6.27)$$

At the point where $Q(\mathbf{r})$ is maximized, r_j and π_j can only be non-zero if j maximizes R_j .

All of this shows that we need only consider $J = 1$ in (6.23). (In case of ties in (6.27) we can arbitrarily choose one of the maximizing j values.) The maximized value of σ_g is then $\sigma_{\max} = (1 - R_{\max})^{-1/2}$ from (6.24) and (6.27), where

$$R_{\max} = \max_{\mu_1} \left\{ \mu_1^2 / \left(1 + c_0 e^{\mu_1^2/2} \right) \right\}. \quad (6.28)$$

The maximizing argument μ_1 ranges from 1.43 for $\pi_0 = 0.95$ to 1.51 for $\pi_0 = 0.70$. This is considerably less than choices such as $\mu_1 = 2.5$, necessary to give small false discovery rates, at which the values in Figure 6.3 will be conservative upper bounds.

6.3 The MLE Method for Empirical Null Estimation

The *MLE method* takes a more straightforward approach to empirical null estimation: starting with the zero assumption (4.44), we obtain normal theory maximum likelihood estimators $(\hat{\delta}_0, \hat{\sigma}_0, \hat{\pi}_0)$ based on the z_i values in

$\mathcal{A}_0$. These tend to be less variable than central matching estimates, though more prone to bias.

Given the full set of z -values $\mathbf{z} = (z_1, z_2, \dots, z_N)$, let N_0 be the number of z_i in $\mathcal{A}_0$ and $\mathcal{I}_0$ their indices,

$$\mathcal{I}_0 = \{i : z_i \in \mathcal{A}_0\} \quad \text{and} \quad N_0 = \#\mathcal{I}_0 \quad (6.29)$$

and define $\mathbf{z}_0$ as the corresponding collection of z -values,

$$\mathbf{z}_0 = \{z_i, i \in \mathcal{I}_0\}. \quad (6.30)$$

Also, let $\varphi_{\delta_0, \sigma_0}(z)$ be the $\mathcal{N}(\delta_0, \sigma_0^2)$ density function

$$\varphi_{\delta_0, \sigma_0}(z) = \frac{1}{\sqrt{2\pi\sigma_0^2}} \exp \left\{ -\frac{1}{2} \left(\frac{z - \delta_0}{\sigma_0} \right)^2 \right\} \quad (6.31)$$

and

$$H_0(\delta_0, \sigma_0) \equiv \int_{\mathcal{A}_0} \varphi_{\delta_0, \sigma_0}(z) dz, \quad (6.32)$$

this being the probability that a $\mathcal{N}(\delta_0, \sigma_0^2)$ variate falls in $\mathcal{A}_0$.

We suppose that the N z_i values independently follow the two-groups model (2.7) with $f_0 \sim \mathcal{N}(\delta_0, \sigma_0^2)$ and $f_1(z) = 0$ for $z \in \mathcal{A}_0$. (In terms of Figure 2.3, $\mathcal{A}_0 = \mathcal{Z}$, $N_0 = N_0(\mathcal{A}_0)$, and $N_1(\mathcal{A}_0) = 0$.) Then $\mathbf{z}_0$ has density and likelihood function

$$f_{\delta_0, \sigma_0, \pi_0}(\mathbf{z}_0) = \left[\binom{N}{N_0} \theta^{N_0} (1 - \theta)^{N - N_0} \right] \left[\prod_{i \in \mathcal{I}_0} \frac{\varphi_{\delta_0, \sigma_0}(z_i)}{H_0(\delta_0, \sigma_0)} \right] \quad (6.33)$$

when

$$\theta = \pi_0 H_0(\delta_0, \sigma_0) = \Pr \{z_i \in \mathcal{A}_0\}. \quad (6.34)$$

Notice that $\mathbf{z}_0$ provides $N_0 = \#\mathbf{z}_0$, distributed as $\text{Bi}(N, \theta)$, while N is a known constant.

Exercise 6.7 Verify (6.33).

Exponential family calculations, described at the end of the section, provide the MLE estimates $(\hat{\delta}_0, \hat{\sigma}_0, \hat{\pi}_0)$. These are usually not overly sensitive to the choice of $\mathcal{A}_0$, which can be made large in order to minimize estimation error. Program `locfdr` centers $\mathcal{A}_0$ at the median of $z_1, z_2, \dots, z_N$, with half-width about twice the preliminary estimate of σ_0 based on interquartile range. (The multiple is less than 2 if N is very large.)

Table 6.3 reports on the results of a small Monte Carlo experiment:

measurements z_i were obtained from model (2.7) with $\pi_0 = 0.95$, $f_0 \sim \mathcal{N}(0, 1)$, $f_1 \sim \mathcal{N}(2.5, 1)$, and $N = 3000$. The z_i values were *not* independent, having root mean square correlation $\alpha = 0.10$; see Chapter 8. One hundred simulations gave means and standard deviations for $(\hat{\delta}_0, \hat{\sigma}_0, \hat{\pi}_0)$ obtained by central matching and by the MLE method.

Table 6.3 *MLE and central matching estimates $(\hat{\delta}_0, \hat{\sigma}_0, \hat{\pi}_0)$ from 100 simulations with $\pi_0 = 0.95$, $f_0 \sim \mathcal{N}(0, 1)$, $f_1 \sim \mathcal{N}(2.5, 1)$, $N = 3000$, and root mean square correlation 0.10 between the z -values. Also shown is correlation between MLE and CM estimates.*

	$\hat{\delta}_0$		$\hat{\sigma}_0$		$\hat{\pi}_0$	
	MLE	CM	MLE	CM	MLE	CM
mean	-.093	-.129	1.004	.984	.975	.963
stdev	.016	.051	.067	.098	.008	.039
corr	.76		.89		.68	

The MLE method yielded smaller standard deviations for all three estimates. It was somewhat more biased, especially for π_0 . Note that each 3000-vector $\mathbf{z}$ had its mean subtracted before the application of `locfdr` so the true null mean δ_0 was -0.125 not 0.

Besides being computationally more stable than central matching of Figure 6.2, the MLE method benefits from using more of the data for estimation, about 94% of the central z -values in our simulation, compared to 50% for central matching. The latter cannot be much increased without some danger of eccentric results. The upward bias of the MLE $\hat{\pi}_0$ estimate has little effect on $\widehat{\text{fdr}}(z)$ or $\widehat{\text{Fdr}}(z)$, but it can produce overly conservative estimates of power (Section 5.4). Taking a chance on more parametric methods (see Notes) may be necessary here.

Straightforward computations produce maximum likelihood estimates $(\hat{\delta}_0, \hat{\sigma}_0, \hat{\pi}_0)$ in (6.33); $f_{\delta_0, \sigma_0, \pi_0}(\mathbf{z}_0)$ is the product of two exponential families⁵ which can be solved separately (the two bracketed terms). The binomial term gives

$$\hat{\theta} = N_0/N \quad (6.35)$$

while $\hat{\delta}_0$ and $\hat{\sigma}_0$ are the MLEs from a truncated normal family, obtained by

⁵ Appendix A gives a brief review of exponential families.

familiar iterative calculations, finally yielding

$$\hat{\pi}_0 = \hat{\theta} / H_0(\hat{\delta}_0, \hat{\sigma}_0) \quad (6.36)$$

from (6.34). The log of (6.33) is concave in $(\delta_0, \sigma_0, \pi_0)$, guaranteeing that the MLE solutions are unique.

Exercise 6.8 Show that (6.33) represents a three-parameter exponential family. That is,

$$\log f_{\delta_0, \sigma_0, \pi_0}(z_0) = \eta_1 Y_1 + \eta_2 Y_2 + \eta_3 Y_3 - \psi(\eta_1, \eta_2, \eta_3) + c(z_0) \quad (6.37)$$

where (η_1, η_2, η_3) are functions of $(\delta_0, \sigma_0, \pi_0)$ and (Y_1, Y_2, Y_3) are functions of z_0 . What are (η_1, η_2, η_3) and (Y_1, Y_2, Y_3) ?

6.4 Why the Theoretical Null May Fail

The four examples in Figure 6.1 strongly suggest failure of the theoretical null distribution $f_0(z) \sim \mathcal{N}(0, 1)$. This is somewhat shocking! Theoretical null derivations like that for Student's t -distribution are gems of the statistical literature, as well as pillars of applied practice. Once alerted, however, it isn't difficult to imagine causes of failure for the theoretical null. The difference in large-scale testing is only that we can detect and correct such failures.

Making use of either central matching or the MLE method, false discovery rate estimates (5.5) and (5.6) now become

$$\widehat{\text{fdr}}(z) = \hat{\pi}_0 \hat{f}_0(z) / \hat{f}(z) \quad \text{and} \quad \widehat{\text{Fdr}}(z) = \hat{\pi}_0 \hat{F}_0(z) / \hat{F}(z) \quad (6.38)$$

with $\hat{F}_0(z)$ the left or right cdf of $\hat{f}_0(z)$ as desired. Statement (6.38) is not a universal improvement over (5.5)–(5.6); estimating the null distribution $f_0(z)$ substantially adds to the variability of false discovery rate estimates, as documented in Chapter 7. But using the theoretical null when it is wrong is a recipe for false inference. What follows is a list of practical reasons why the theoretical null might go astray.

(I) *Failed mathematical assumptions* Textbook null hypothesis derivations usually begin from an idealized mathematical framework, e.g., independent and identically distributed (*i.i.d.*) normal components for a two-sample t -statistic. Deviations from the ideal can be easy to spot in large-scale data sets. For example, the logged expression levels x_{ij} for the HIV data have longer-than-normal tails. (Although, by itself, that is not enough to induce the underdispersion seen in the z -value histogram of Figure 6.1d:

repeating the HIV computation for a 7680×8 matrix whose components were randomly selected from the actual matrix gave an almost perfectly $N(0, 1)$ z -value histogram.)

(II) *Correlation across sampling units* Student's theoretical null distribution for a two-sample t -statistic (2.2) assumes independence across the n sampling units: for instance, across the 72 patient scores for any one gene in the leukemia study. Chapter 8 shows how such independence assumptions can fail in practice. In large-scale studies, even minor experimental defects can manifest themselves as correlation across sampling units. The expression levels x_{ij} for the prostate study of Section 2.1 will be seen to have drifted slightly as the experiment went on (i.e., as j increased), some genes drifting up and others down, inducing minor correlations across microarrays. Other data sets will show more significant correlations, big enough to seriously distort the z -value histogram.

(III) *Correlation across cases* It was argued in Section 4.4 that independence among the z -values is *not* required for valid false discovery rate inference. The hitch is that this is only true if we are using the correct null distribution $f_0(z)$. Section 8.3 discusses the following disconcerting fact: even if the theoretical null distribution $z_i \sim N(0, 1)$ is valid for all null cases, correlation among the z_i can make $N(0, 1)$ a misleading choice in likely realizations of $\mathbf{z} = (z_1, z_2, \dots, z_N)$.

Figure 6.4 provides an example. A simulation study was run with $N = 6000$ z -values; 5700 were null cases having $z_i \sim N(0, 1)$, with root mean square correlation 0.1 among the $5700 \cdot 5699/2$ pairs. The 300 non-null cases followed an exact $N(2.5, 1)$ distribution, that is,

$$z_i = 2.5 + \Phi^{-1}\left(\frac{i - 0.5}{300}\right) \quad \text{for } i = 1, 2, \dots, 300. \quad (6.39)$$

Three quantities were calculated for each of 100 simulations,

$$\text{Fdp}(2.5) = \frac{\#\{\text{null } z_i \geq 2.5\}}{\#\{z_i \geq 2.5\}}, \quad (6.40)$$

the actual false discovery proportion (2.28) for $\mathcal{Z} = [2.5, \infty)$, and both $\widehat{\text{Fdr}}_{\text{theo}}(2.5)$ and $\widehat{\text{Fdr}}_{\text{emp}}(2.5)$, the theoretical and empirical null Fdr estimates (5.6) and (6.38).

The two Fdr estimates are plotted versus Fdp in Figure 6.4. Correlation between the null cases makes $\text{Fdp}(2.5)$ highly variable. $\widehat{\text{Fdr}}_{\text{emp}}(2.5)$ follows $\text{Fdp}(2.5)$, somewhat noisily, but $\widehat{\text{Fdr}}_{\text{theo}}(2.5)$ moves in the wrong direction, slowly decreasing as Fdp increases. $\widehat{\text{Fdr}}_{\text{emp}}$ is much more vari-

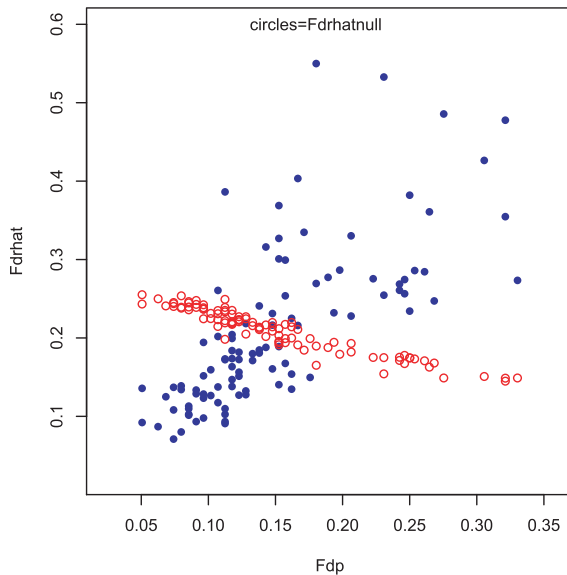


Figure 6.4 Simulation experiment comparing empirical null and theoretical null estimates of $Fdr(2.5)$, plotted against actual false discovery proportion $Fdp(2.5)$, as described in the text. The theoretical null estimates (open circles) decrease as Fdp increases.

able than $\widehat{Fdr}_{theo}$ but nevertheless more accurate as an estimate of Fdp , with mean $|\widehat{Fdr}_{emp}(2.5) - Fdp(2.5)| = 0.065$ compared to mean $|\widehat{Fdr}_{theo}(2.5) - Fdp(2.5)| = 0.098$. Even though the theoretical $N(0, 1)$ null is unconditionally correct here, it is unsatisfactory conditionally.

Exercise 6.9 Explain why $\widehat{Fdr}_{theo}(2.5)$ is a decreasing function of $Fdp(2.5)$.

(IV) *Unobserved covariates* Except for the chi-square data, all of our examples are observational studies. The 72 leukemia patients were observed, not assigned, to be in the AML or ALL class. Unobserved covariates such as age, gender, concomitant health conditions, processing variables,⁶ etc., may affect the AML/ALL comparison. If the covariates were available we could use linear model techniques to account for them, but if not they tend to broaden the effective null distribution f_0 , as discussed next.

Suppose we have a microarray experiment comparing $n/2$ Class A sub-

⁶ For example, different areas on a microarray chip are read separately, by devices that may be calibrated differently.

jects with $n/2$ Class B subjects, for N genes. The null genes are assumed to have expression levels

$$x_{ij} = u_{ij} + \frac{I_j}{2}\beta_i \quad \begin{cases} u_{ij} \sim \mathcal{N}(0, 1) \\ \beta_i \sim \mathcal{N}(0, \sigma_\beta^2) \end{cases} \quad (6.41)$$

with $u_{i1}, u_{i2}, \dots, u_{in}, \beta_i$ mutually independent, and

$$I_j = \begin{cases} -1 & j = 1, 2, \dots, n/2 \\ 1 & j = n/2 + 1, \dots, n. \end{cases} \quad (6.42)$$

Here the β_i values are small disturbances caused by unequal effects of unobserved covariates on the two classes. (The β_i may be correlated across genes.)

It is easy to show that the two-sample t -statistic t_i (2.2) comparing Class A with Class B follows a dilated t -distribution with $n - 2$ degrees of freedom,

$$t_i \sim \left(1 + \frac{n}{4}\sigma_\beta^2\right)^{\frac{1}{2}} \cdot t_{n-2} \quad (6.43)$$

for the null cases. In other words, the null density is $(1 + n\sigma_\beta^2/4)^{1/2}$ times more dispersed than the usual theoretical null.

Exercise 6.10 (a) Verify (6.43). (b) Suppose that all the genes are null and that $\sigma_\beta = 2/\sqrt{n}$. Show that, for large N , the local fdr, with $\pi_0 = 1$, will be about

$$\text{fdr}(t_i) \doteq \sqrt{2}f_{n-2}(t_i) / f_{n-2}(t_i / \sqrt{2}) \quad (6.44)$$

when f_0 is taken to be the usual theoretical null $f_{n-2}(t)$, a Student t -density with $n - 2$ degrees of freedom. (c) For $n = 20$, what is the probability that $\text{fdr}(t_i) \leq 0.20$ under (6.44)?

The empirical null fdr estimate $\hat{\pi}_0 \hat{f}_0(z) / \hat{f}(z)$ scales “correctly” using either central matching or the MLE method: if each z_i is multiplied by the same positive constant c , the value of $\widehat{\text{fdr}}$ stays the same for all cases. Another way to say this is that our interest in an outlying z_i is judged relative to the width of the histogram’s center, rather than relative to the theoretical $\mathcal{N}(0, 1)$ width.

Unobserved covariates are ubiquitous in large-scale studies and are perhaps the most common source of trouble for the theoretical null, a likely culprit for the kind of gross overdispersion seen in Figure 6.1a. Reason (III) above, correlation across cases, produces smaller effects, but is capable of

causing the underdispersion seen in Figure 6.1d as well as overdispersion (Chapter 7). Microarray studies are particularly prone to correlation effects, as the examples of Chapter 8 will show. Reason (II), correlation across the supposedly independent sampling units, is surprisingly prevalent as a possible source of overdispersion (Chapter 8). Failed mathematical assumptions, Reason (I), is the only one of the four failure causes that is easily cured by permutation calculations, as discussed in the next section.

Our list of causes is by no means complete. *Filtration*, the data-based preselection of a subset of promising-looking cases for final analysis, can distort both the null and non-null distributions. In a microarray context the investigator might first select only those genes having above-average standard deviations, on the grounds that this hints at differential response across the various experimental conditions. Filtration is a dangerous tactic. Among other dangers, it reduces, in a biased manner, information available for evaluating the appropriate null distribution.

6.5 Permutation Null Distributions

Permutation techniques for assessing a null distribution lie somewhere between the theoretical and empirical methods, but closer to the former than the latter. They are easiest to describe in two-sample situations like the prostate, leukemia, and HIV studies. We have an $N \times n$ data matrix $\mathbf{X}$ with the first n_1 columns representing Treatment 1 and the last n_2 columns Treatment 2. These have produced an N -vector $\mathbf{z}$ of z -values, perhaps as in (2.2)–(2.5).

In order to compute the permutation null distribution, we randomly permute the columns of $\mathbf{X}$ as in (3.39) and recalculate the vector of z -values. Repeating the process B times gives vectors $\mathbf{z}^{*1}, \mathbf{z}^{*2}, \dots, \mathbf{z}^{*B}$, with

$$\mathbf{Z}^* = (\mathbf{z}^{*1}, \mathbf{z}^{*2}, \dots, \mathbf{z}^{*B}) \quad (6.45)$$

representing the full $N \times B$ matrix of permuted values. The usual permutation null is then

$$\hat{f}_0^{\text{perm}} = \text{the empirical distribution of all } N \cdot B \text{ values } z_i^{*b}. \quad (6.46)$$

(Less commonly, we might calculate a separate permutation null for each case i from the B values in the i th row of $\mathbf{Z}^*$. This requires B to be very large in order to assess the extreme tail probabilities necessary in large-scale testing, whereas $B = 10$ or 20 is often sufficient for (6.46).)

Figure 6.5 shows QQ plots⁷ of $\hat{f}_0^{\text{perm}}$ for the leukemia and HIV studies

⁷ A QQ plot of observations $x_1, x_2, \dots, x_m$ plots the i th ordered value $x_{(i)}$ versus the

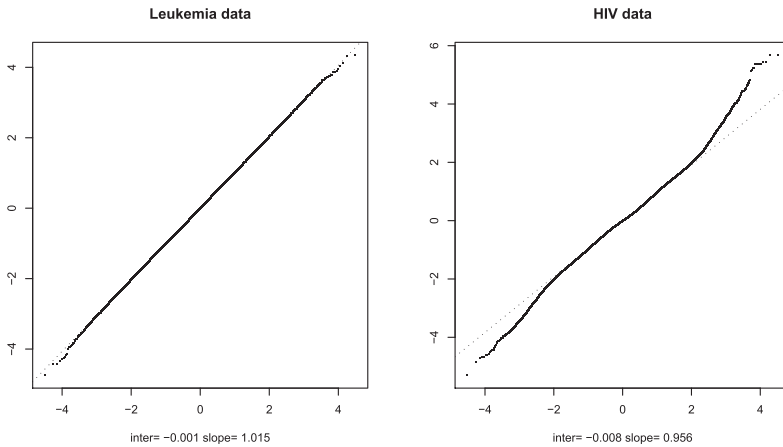


Figure 6.5 *QQ* plots of permutation null distributions for the leukemia and HIV studies, $B = 20$ permutations each. Dotted lines are least squares fits to central 80% of the plot, with intercepts and slopes as indicated; $\hat{f}_0^{\text{perm}} \sim \mathcal{N}(0, 1)$ for leukemia study but has heavier tails for HIV study.

(z -values as in (2.2)–(2.5)), in both cases based on $B = 20$ permutations. The leukemia plot shows $\hat{f}_0^{\text{perm}}$ nearly $\mathcal{N}(0, 1)$ while $\hat{f}_0^{\text{perm}}(z)$ for the HIV data is roughly $\mathcal{N}(0, 0.96^2)$ near $z = 0$, but with heavier-than-normal tails.

Permutation methods offer a simple and attractive way to deal with mathematically complicated hypothesis testing situations. They are not however a remedy for the theoretical null failures seen in Figure 6.1. There are several points to consider:

- Of the four potential causes of failure raised in Section 6.4, permutation methods deal most effectively with Reason (I), failed mathematical assumptions. The permutation process enforces an i.i.d. structure, so that $\hat{f}_0^{\text{perm}}$ is relevant only in that context. (The second i , “identical,” may not be realistic, but this usually doesn’t seem to cause major problems.) Non-standard statistical definitions — for example, adding a constant to the usual t -statistic denominator, as in SAM — are automatically incorporated into the permutation calculations.
- Permutation methods are no help with Reason (II), correlation across

corresponding normal quantile $\Phi^{-1}[(i - 0.5)/m]$. If the x values come from a $\mathcal{N}(a, b^2)$ distribution, the plot will approximate a straight line with intercept a and slope b .

sampling units, since the sampling process effectively enforces independence.⁸

- They are also of no help with Reason (IV), unobserved covariates. The data vector $(x_{i1}, x_{i2}, \dots, x_{in})$ in (6.41) has $\beta_i/2$ subtracted from the first $n/2$ observations and added to the last $n/2$. But as far as permutation samples are concerned, the $\pm\beta_i/2$ disturbances are randomly distributed across the n observations. The permutation distribution of the two-sample t -statistic is nearly t_{n-2} under model (6.41), no matter how large β_i may be, rather than equaling the dilated distribution (6.43).
- A virtue of permutation methods is that they preserve correlation between cases, e.g., between genes (since the columns of $\mathbf{X}$ are permuted intact). We can estimate the correlation between any pair of z -values from the correlation between the rows of $\mathbf{Z}^*$ (6.45), as was done in Section 3.4 and will be seen again in Chapter 8.

Nevertheless, permutation methods are of no direct assistance with Reason (III), the effects of between-case correlations on the proper choice of f_0 . For any given row i of $\mathbf{Z}^*$, the values z_i^{*b} do not depend on between-case correlations, and neither does $\hat{f}_0^{\text{perm}}$. Both the leukemia and HIV examples display considerable case-wise correlation (see Chapter 8) but still have $\hat{f}_0^{\text{perm}}$ close to the theoretical null and far from the empirical null.

- In fact, $\hat{f}_0^{\text{perm}}$ will *usually* approximate a $\mathcal{N}(0, 1)$ distribution for (2.5), the z -score version of the t -statistic. Fisher's introduction of permutation arguments was intended to justify Student's distribution in just this way. A considerable body of theory in the 1950s showed $\hat{f}_0^{\text{perm}}$ converging quickly to $\mathcal{N}(0, 1)$ as n grew large. The results in Figure 6.5 are typical, showing almost perfect convergence in the leukemia example, $n = 72$.
- Permutation and empirical methods can be combined by letting the permutation null replace $\mathcal{N}(0, 1)$ in the empirical algorithm. That is, we perform the empirical null calculations on $\Phi^{-1}(F_{\text{perm}}(z_i))$ rather than on the original z -values, with F_{perm} the permutation cdf. Doing so made no difference to the leukemia data but considerably increased $\widehat{\text{fdr}}$ for the HIV data, reducing the number of cases with $\widehat{\text{fdr}} \leq 0.20$ from 128 to 42, now all on the left side.
- Permutation methods are not restricted to two-sample problems. They depend on symmetries of the null hypothesis situation, which may or

⁸ It actually enforces a small negative correlation: if x_1^* and x_2^* are any two draws in a permutation sample, then their correlation is $-1/(n-1)$ with respect to the permutation process, regardless of the correlations in the original sample $x_1, x_2, \dots, x_n$.

may not exist. They would be difficult to find, perhaps impossible, for the chi-square and police examples of Figure 6.1. Bootstrap methods are more flexible but have their own problems in large-scale situations; see Section 7.5.

Notes

The leukemia data was the motivating example in Golub et al. (1999), an early exploration of advanced statistical approaches to microarray experiments. Ridgeway and MacDonald (2009) discuss the police data, providing a much more extensive analysis than the one here. Van't Wout et al. (2003) originated the HIV data, which was used as a main example in Gottardo et al. (2006) as well as Efron (2007b). The chi-square data is part of an ongoing unpublished experiment.

Empirical null methods were introduced in Efron (2004), along with the central matching algorithm. Efron (2007b) suggested the MLE method. Why the theoretical null might fail is discussed in Efron (2008a). Jin and Cai (2007, 2010) present a characteristic function approach to estimating the empirical null, while a more parametric normal mixture method is analyzed in Muralidharan (2010).

The 1935 edition of Fisher's *The Design of Experiments* introduced permutation methods as a justification of Student's *t*-test. A great deal of subsequent analysis by many authors, including Pitman (1937) and Hoeffding (1952), supported Fisher's contention that permutation methods usually agree closely with Student's distribution.